Functional Requirements Document – Application Settings

TaskFlow – Intelligent Task Allocation & Tracking System

> **Document:** Functional Requirements Document · **Version:** 1.0 · **Status:** Draft
> **Project type:** Personal Project (Skill Development) · **Date:** 31-Oct-2026
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> **Based on:** BRD-Application-Settings.md v1.0
> **Module:** Settings · **Sub-module:** 06.2 Application Settings · **Code:** APP
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Document Control

Revision History

Version	Date	Author	Summary of changes
1.0	31-Oct-2026	Jojo George	Initial version

Related Documents

Document	Purpose	Location
BRD – Application Settings	Business needs this FRD details (REQ-APP-nn, BR-APP-nn, AC-APP-nn)	`../../../BRD/06-Settings/06.2-Application-Settings/BRD-Application-Settings.md`
Overall FRD	Roles, global rules, Authentication, cross-cutting behaviour	`../../00-Overall/FRD-TaskFlow-Overall.md`
FRD – Reminder Settings	Reminder defaults and Mail delivery; shows this timezone read-only	`../06.1-Reminder-Settings/FRD-Reminder-Settings.md`
FRD – Notification	Bell notifications controlled by the in-app switch	`../../04-Communication/04.2-Notification/FRD-Notification.md`
FRD – Task Management	Create Task, which pre-selects the default priority	`../../02-TaskFlow-Monitoring/02.1-Task-Management/FRD-Task-Management.md`
Development Plan	Screen-wise build plan (Screen 20)	`TF - Documents/Screens Wise Feature Development Plan/20-application-settings-screen.md`
HLD / LLD	Technical design	`../../../HLD/06-Settings/06.2-Application-Settings/HLD-Application-Settings.md`, `../../../LLD/06-Settings/06.2-Application-Settings/LLD-Application-Settings.md`

1. Introduction

1.1 Purpose

This FRD specifies the detailed, testable behaviour of the Application Settings screen and of the organization-wide values it maintains: the application name, the application timezone, the default task priority and the in-app notifications switch – including how other screens read and use them.

1.2 Scope

In scope: viewing and changing the four values; validation and saving; showing the name in the application shell; using the timezone for every "today" decision; pre-selecting the default priority on Create Task; pausing bell notifications; the link to Reminder Settings; reading of the non-secret values by every signed-in user.

Out of scope (BRD §3.2): reminder intervals and Mail delivery (Reminder Settings); AI provider configuration (TFA Settings); a full change history; reset to factory defaults; outbound SMTP settings (superseded by internal Mail).

1.3 Audience

Developer, tester, reviewer and product owner of TaskFlow.

1.4 Conventions

- Requirement IDs: `FR-APP-nn`. Use cases: `UC-APP-nn`. Business rules: `BR-APP-nn` (sub-module BRD) and `BR-nn` (overall BRD).
- Priority: **Must / Should / Could** (MoSCoW).
- "The system shall ..." = mandatory behaviour; "should" = recommended.
- "BRD trace" refers to `REQ-APP-nn` in the Application Settings BRD.

1.5 Definitions & Abbreviations

Term	Meaning
Application name	Brand name shown in the sidebar and header (default "TaskFlow")
Application timezone	The one timezone (for example `Asia/Kolkata`) that decides which calendar day a deadline falls on
Default task priority	Priority pre-selected on Create Task: Low, Medium, High or Critical
In-app notifications	Bell notifications shown in the Notification Center
Configured timezone	The timezone set in the server configuration by whoever installs TaskFlow; first value and fallback

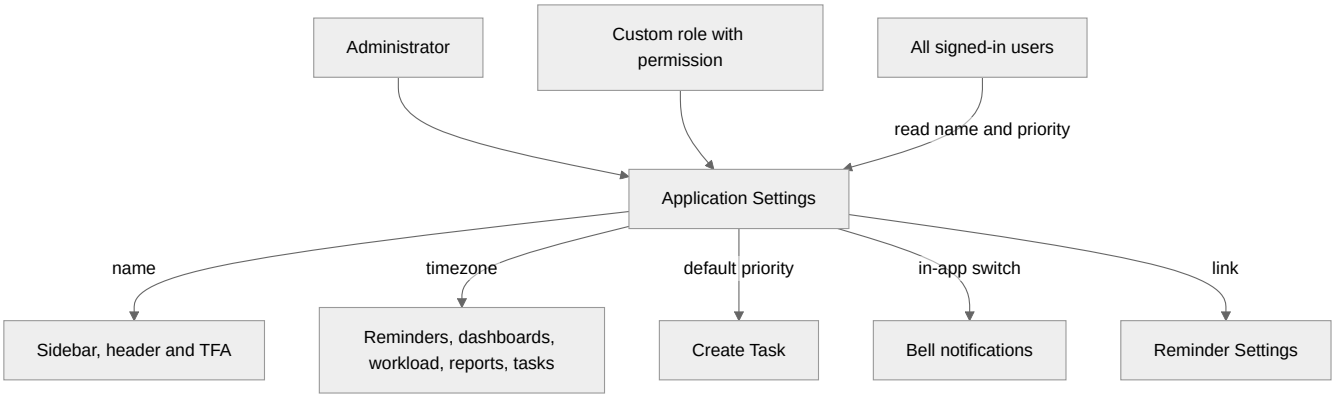
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2. System Overview

2.1 Module Map

Code	Module	Child modules	Primary roles
APP	Settings → Application Settings	Application Settings (one screen)	Administrator, Custom role with `settings.application.manage`
REM	Settings → Reminder Settings (related)	Reminder defaults, Mail delivery	Administrator, Manager
NOTIF	Communication → Notification (related)	Notification Center, reminder engine	All users
TASK	Monitoring → Task Management (related)	Create Task	Manager, Administrator

2.2 Context Diagram



3. User Roles & Permissions

3.1 Roles

Role	Description	Data scope
Administrator	Views and changes application settings	Organization
Manager	No access to the screen by default; sees the name and gets the default priority on Create Task	Organization (read behind the scenes)
Employee	No access to the screen by default; sees the name	Organization (read behind the scenes)
Custom role	Full screen when `settings.application.manage` is granted; otherwise reads behind the scenes only	Organization

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3.2 Permission Matrix (functional view)

Module › Function	Permission key	Administrator	Manager	Employee	Custom role
Settings › Application Settings › Open screen and change settings	`settings.application.manage`	Granted	Denied	Denied	Configurable
Application shell, Create Task › Read application name and default priority	– (signed in)	Always On	Always On	Always On	Always On

> Manager and Employee values are the defaults given when the permission is first created. The Administrator role is given every permission again each time the application starts.

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4. Functional Requirements by Module

4.1 Module: Application Settings (APP)

Purpose: Maintain the organization-wide basic configuration of TaskFlow. · **BRD reference:** BRD-Application-Settings SCREEN 1, BR-APP-01…BR-APP-06 (original BRD Screen 20 "System Settings")

4.1.1 Screen: Application Settings

Field	Value
Route / location	`/settings/application` (menu **Settings → Application Settings**)
Access	Administrator; Custom role with `settings.application.manage`
Entry points	Sidebar menu item **Application Settings**

Screen elements

Element	Type	Description	Mandatory	Visible to
Application settings	Page title	With the line "Application-wide configuration: the brand name, timezone, default task priority and in-app notifications."	–	Holders of `settings.application.manage`
Application name	Text input	Hint: "Shown in the sidebar and header."	Yes	Same
Default timezone	Text input with suggestions	Suggestions UTC, Asia/Kolkata, Asia/Dubai, Europe/London, Europe/Berlin, America/New_York, America/Chicago, America/Los_Angeles, Australia/Sydney; any valid timezone name can be typed. Hint: "Decides which calendar day a deadline belongs to. Applied immediately, no restart needed."	Yes	Same
Default task priority	Drop-down	Low, Medium, High, Critical. Hint: "Pre-selected on the Create Task form."	Yes	Same
Enable in-app notifications	Switch	Hint: "When off, no new bell notifications are created for any event. Existing notifications are kept."	Yes	Same

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Element	Type	Description	Mandatory	Visible to
Reminder defaults	Link	Opens **Reminder Settings**, where reminder intervals and Mail delivery are kept	–	Same
Save settings	Button	Saves all four values; shows "Saving…" while saving	–	Same
Saved	Tick	Shown after a successful save	–	Same
Loading placeholder / error card with **Retry**	Placeholder, card + button	While loading / when loading fails	–	Same

Functional requirements

ID	Requirement	Priority	Permission	BRD trace
FR-APP-01	The system shall load and display the current application name, timezone, default task priority and in-app notifications switch when the screen opens.	Must	`settings.application.manage`	REQ-APP-01
FR-APP-02	The system shall show a loading placeholder while loading and, if loading fails, the error card "Could not load the application settings." (or the server's message) with a **Retry** button that loads again.	Should	`settings.application.manage`	REQ-APP-01
FR-APP-03	The system shall let the user change **Application name**.	Must	`settings.application.manage`	REQ-APP-02
FR-APP-04	The system shall require an application name of 1–100 characters after removing spaces before and after it, and shall save the trimmed name.	Must	`settings.application.manage`	REQ-APP-03
FR-APP-05	The system shall show the saved application name in the sidebar and header straight after a successful save, without a page reload, and to every other signed-in user.	Must	– (signed in)	REQ-APP-04
FR-APP-06	The system shall offer timezone suggestions in **Default timezone** and accept any typed timezone name the server recognises.	Must	`settings.application.manage`	REQ-APP-05
FR-APP-07	The system shall refuse a blank timezone ("Timezone is required." on the screen; "Timezone cannot be blank." when saving) and an unrecognised one ("Unknown timezone.").	Must	`settings.application.manage`	REQ-APP-06
FR-APP-08	The system shall use the saved timezone for every "today", "due today" and "overdue" decision – reminders, dashboards, workload, reports and task deadline checks – straight after saving, without a restart.	Must	– (system)	REQ-APP-07, BR-APP-02
FR-APP-09	The system shall use UTC, and record the problem in the server log, when the saved timezone cannot be recognised at run time, so that reminders and dashboards keep working.	Must	– (system)	REQ-APP-08
FR-APP-10	The system shall let the user choose **Default task priority** as Low, Medium, High or Critical.	Must	`settings.application.manage`	REQ-APP-09

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ID	Requirement	Priority	Permission	BRD trace
FR-APP-11	The system shall pre-select the saved default task priority on the Create Task form.	Must	– (Create Task)	REQ-APP-10
FR-APP-12	The system shall let the user switch **Enable in-app notifications** on or off for the whole organization.	Must	`settings.application.manage`	REQ-APP-11, BR-APP-01
FR-APP-13	The system shall create no new bell notification for any event while in-app notifications are off, and shall keep all existing notifications.	Must	– (system)	REQ-APP-12, BR-APP-05
FR-APP-14	The system shall show in the **Reminder defaults** group a link that opens Reminder Settings, and shall not offer reminder intervals or Mail delivery on this screen.	Should	`settings.application.manage`	REQ-APP-13, BR-APP-06
FR-APP-15	The system shall check every value again when a save is received, independently of the screen (name, timezone, priority).	Must	`settings.application.manage`	REQ-APP-14
FR-APP-16	The system shall refuse a save in which any of the four values is missing, without filling it with a default.	Must	`settings.application.manage`	REQ-APP-15, BR-APP-03
FR-APP-17	The system shall use name "TaskFlow", priority Medium, in-app notifications on and the configured timezone (or UTC when none is configured) for any value that is missing or unreadable, and shall create missing values at start-up without overwriting saved ones.	Must	– (system)	REQ-APP-16
FR-APP-18	The system shall let every signed-in user read the application settings, for the brand name and the default priority.	Must	– (signed in)	REQ-APP-17, BR-APP-04
FR-APP-19	The system shall return only the application name, timezone, default priority and in-app switch to readers, and no secret or sensitive configuration.	Must	– (signed in)	REQ-APP-18
FR-APP-20	The system shall record the user who last saved the application settings and the date and time of the save.	Should	`settings.application.manage`	REQ-APP-19
FR-APP-21	The system shall show "Saving..." on the button while saving, confirm a successful save with the toast "Application settings saved" and the **Saved** tick, and show "Could not save the application settings." when the save fails.	Should	`settings.application.manage`	REQ-APP-20
FR-APP-22	The system shall hide the **Application Settings** menu item from users without `settings.application.manage` and return such users to their dashboard when they open `/settings/application`.	Must	`settings.application.manage`	REQ-APP-21

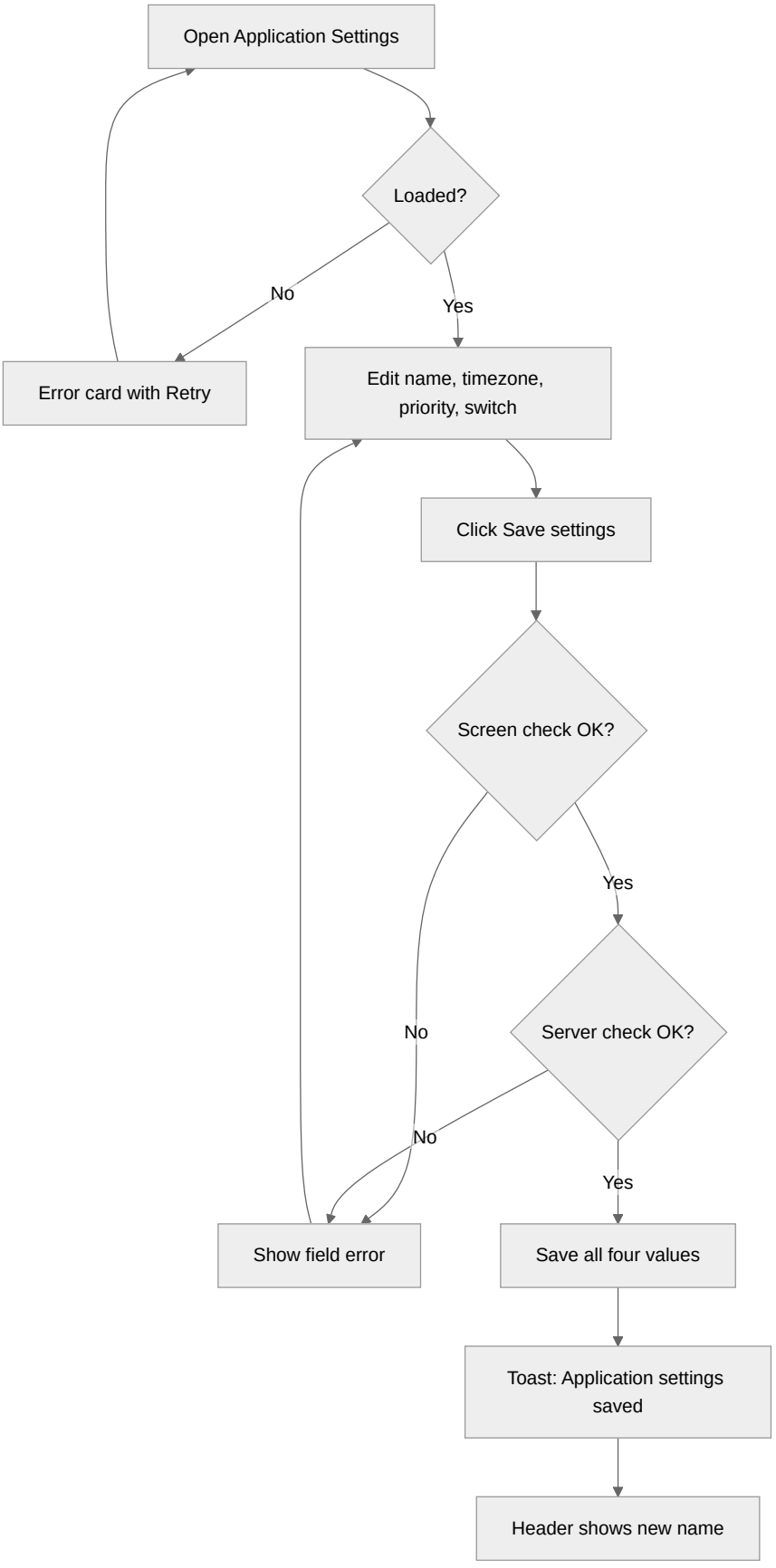
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ID	Requirement	Priority	Permission	BRD trace
FR-APP-23	The system shall refuse every save of application settings requested by a user without `settings.application.manage`.	Must	`settings.application.manage`	REQ-APP-22, BR-APP-04

Use case: UC-APP-01 – Change and save application settings

Item	Detail
Actor	Administrator or Custom role user
Preconditions	Signed in; holds `settings.application.manage`
Trigger	Opens **Settings → Application Settings**
Main flow	1. System shows the current four values (FR-APP-01). 2. User changes the name, timezone, default priority and/or the in-app switch. 3. User clicks **Save settings**; the button shows "Saving…". 4. The screen checks name and timezone (FR-APP-04, FR-APP-07). 5. System checks all values again (FR-APP-15, FR-APP-16). 6. System saves all four values and records editor and time (FR-APP-20). 7. Toast "Application settings saved" and **Saved** tick (FR-APP-21). 8. The new name appears in the sidebar and header at once (FR-APP-05); the new timezone is used at once (FR-APP-08).
Alternate flows	A1: Loading fails → error card with **Retry** → step 1. A2: User follows the **Reminder Settings** link (FR-APP-14).
Exceptions	E1: Name blank or over 100 characters → "Application name is required and must be 100 characters or fewer." E2: Timezone blank → "Timezone is required." E3: Timezone unrecognised → "Unknown timezone." E4: Value missing / invalid priority → save refused. E5: Save fails → "Could not save the application settings."
Postconditions	Four values saved together; last editor and time recorded.

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Use case: UC-APP-02 – Use the application timezone for "today"

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Item	Detail
Actor	System (reminder engine, dashboards, workload, reports, tasks)
Preconditions	A timezone is saved or configured
Trigger	Any calculation needs the current calendar day
Main flow	1. System reads the saved timezone. 2. System converts the current time into it. 3. System decides "due today" and "overdue" by that calendar day (FR-APP-08).
Alternate flows	A1: No timezone saved → configured timezone, or UTC (FR-APP-17).
Exceptions	E1: Saved timezone not recognised on the server → problem recorded, UTC used until a valid timezone is saved (FR-APP-09).
Postconditions	Every screen gives the same answer for "due today" and "overdue".

Use case: UC-APP-03 – Pause bell notifications

Item	Detail
Actor	Administrator; System
Preconditions	Signed in; holds `settings.application.manage`
Trigger	The organization wants no new bell notifications for a period
Main flow	1. User switches **Enable in-app notifications** off and saves (UC-APP-01). 2. For every later event, the system creates no bell notification (FR-APP-13). 3. Existing notifications stay in the Notification Center. 4. Switching it on again resumes new bell notifications.
Alternate flows	–
Exceptions	E1: Value unreadable → treated as on (FR-APP-17).
Postconditions	No new bell notifications while off; Mail copies of reminders follow the Reminder Settings switch.

Field validations

Field	Rule	Error message
Application name	Required; 1–100 characters after trimming	Screen: "Application name is required and must be 100 characters or fewer." Server: "Application name cannot be blank."
Default timezone	Required; must be a timezone name the server recognises	Screen: "Timezone is required." Server: "Timezone cannot be blank." / "Unknown timezone."
Default task priority	Required; Low, Medium, High or Critical	Save refused
Enable in-app notifications	Required; on or off	Save refused when missing

Business rules applied: BR-APP-01, BR-APP-02, BR-APP-03, BR-APP-04, BR-APP-05, BR-APP-06; global rules of the overall BRD: one consistent timezone for deadline comparisons; consistent calculation rules for dashboards and reports; in-app notifications work independently of other delivery channels.

Notifications / side effects

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Event	Who is notified	Channel
Settings saved	Saving user	Toast "Application settings saved"
Name changed	Every user	Sidebar and header (and the TFA brand name)
In-app switch off	All users	No new bell notifications; existing kept
Timezone changed	–	Reminders, dashboards, workload, reports, tasks use it at once

Messages

Situation	Message
Success	"Application settings saved" (toast) and **Saved** tick
Name blank or too long (screen)	"Application name is required and must be 100 characters or fewer."
Name blank (server)	"Application name cannot be blank."
Timezone blank (screen / server)	"Timezone is required." / "Timezone cannot be blank."
Timezone not recognised	"Unknown timezone."
Load failed	"Could not load the application settings." with **Retry**
Save failed	"Could not save the application settings."
No permission	Menu item hidden; opening `/settings/application` returns the user to their dashboard; a direct save is refused

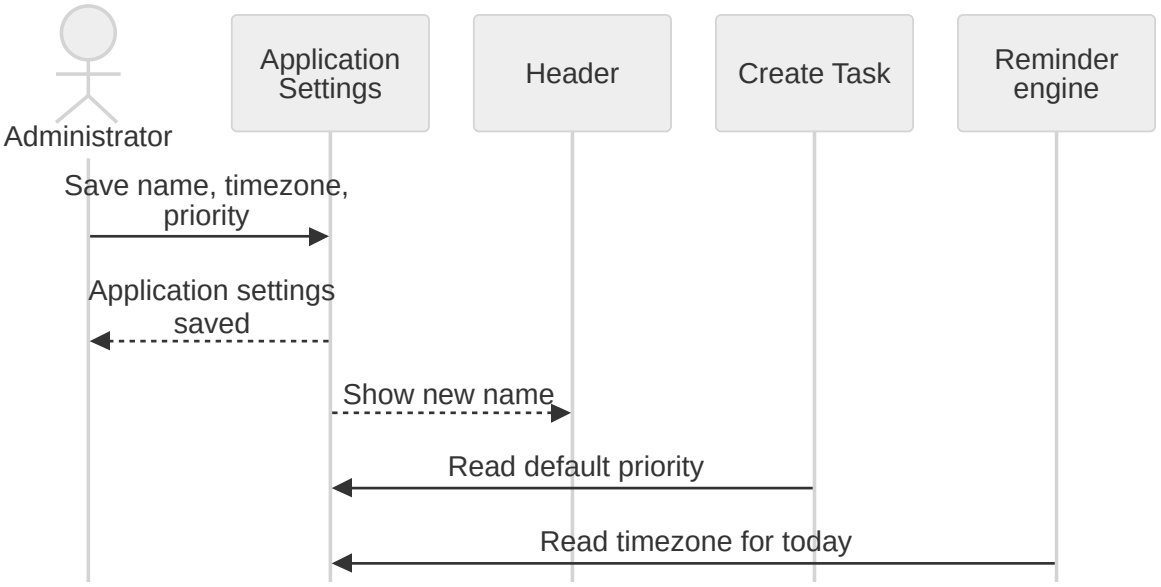
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5. Cross-Module Workflows

5.1 Rename and re-zone the application

Step	Module	Actor / System	Action	FR reference
1	APP	Administrator	Saves "Acme TaskFlow", `Asia/Dubai`, priority High	FR-APP-03...FR-APP-10, FR-APP-15
2	APP	System	Shows "Acme TaskFlow" in sidebar and header	FR-APP-05
3	TASK	Manager	Opens Create Task; High is pre-selected	FR-APP-11
4	NOTIF	System	Next reminder run decides the deadline day in `Asia/Dubai`	FR-APP-08
5	REM	Authorized user	Sees `Asia/Dubai` read-only on Reminder Settings	Reminder Settings FRD

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5.2 In-app switch and notifications

Step	Module	Actor / System	Action	FR reference
1	APP	Administrator	Switches **Enable in-app notifications** off	FR-APP-12
2	NOTIF	System	Checks the switch before creating any bell notification; creates none	FR-APP-13
3	MAIL	System	Reminder copies still follow Reminder Settings Mail delivery	Reminder Settings FRD

6. Reports & Exports

Not applicable for this sub-module. Reports use the application timezone (Reports FRD).

7. Notifications & Messages Catalogue

Code	Trigger	Recipients	Channel	Text
N-APP-01	Successful save	Saving user	Toast	"Application settings saved"
N-APP-02	Name invalid (screen)	Saving user	Inline	"Application name is required and must be 100 characters or fewer."
N-APP-03	Name blank (server)	Saving user	Inline	"Application name cannot be blank."
N-APP-04	Timezone blank	Saving user	Inline	"Timezone is required." / "Timezone cannot be blank."
N-APP-05	Timezone unknown	Saving user	Inline	"Unknown timezone."

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Code	Trigger	Recipients	Channel	Text
N-APP-06	Load failed	Current user	Error card	"Could not load the application settings."
N-APP-07	Save failed	Saving user	Toast / inline	"Could not save the application settings."

8. Data Requirements (functional view)

Data item	Description	Created by	Rules
Application name	Brand name	System at start-up (default "TaskFlow"); changed by user	1–100 characters, trimmed
Application timezone	Timezone for "today"	System at start-up (configured timezone); changed by user	Must be recognised; UTC used at run time if not
Default task priority	Pre-selection on Create Task	System at start-up (Medium); changed by user	Low, Medium, High or Critical
In-app notifications	Bell notifications on/off	System at start-up (on); changed by user	Missing or unreadable counts as on
Last changed by / at	Editor and time of the last save	System on save	No history of earlier values

9. Non-Functional Requirements (functional impact)

ID	Requirement	Affects
NFR-APP-01	The screen loads and saves without noticeable delay	Application Settings
NFR-APP-02	Readable settings never contain secrets	FR-APP-18, FR-APP-19
NFR-APP-03	Changes are checked on every request	FR-APP-15, FR-APP-23
NFR-APP-04	A bad or missing value never stops the application	FR-APP-09, FR-APP-17
NFR-APP-05	Changes are visible immediately (name without reload, timezone without restart)	FR-APP-05, FR-APP-08
NFR-APP-06	Works in current desktop browsers, responsive layout	Application Settings
NFR-APP-07	Every field has a label and hint; errors appear next to the field	Screen elements, validations

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10. Assumptions & Open Issues

Type	Description	Owner
Assumption	One organization per installation, so one set of application settings.	Jojo George
Assumption	The server knows standard timezone names such as `Asia/Kolkata` and `UTC`.	Jojo George
Open issue	⚠️ To be confirmed: whether a full change history of settings (original Screen 20 FR-6) is still required; only the last editor and time are kept.	Jojo George
Open issue	⚠️ To be confirmed: the exact message returned for an invalid priority or a missing value (standard validation wording is used).	Jojo George

Change from original BRD

Original BRD item	Current requirement
Screen 20 "Default task reminder intervals" / FR-3	Kept on Reminder Settings; this screen only links to it (FR-APP-14).
Screen 20 "Notification preferences"	One organization-wide switch **Enable in-app notifications** (FR-APP-12, FR-APP-13).
Screen 20 FR-6 "Record significant configuration changes"	Last editor and time recorded (FR-APP-20); no history.
Screen 20 access "Administrator"	Controlled by `settings.application.manage` (Administrator by default).

11. Requirements Traceability Matrix

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-APP-01	FR-APP-01, FR-APP-02	Application Settings	UC-APP-01	AC-APP-01
REQ-APP-02	FR-APP-03	Application Settings	UC-APP-01	AC-APP-02
REQ-APP-03	FR-APP-04	Application Settings	UC-APP-01	AC-APP-03
REQ-APP-04	FR-APP-05	Sidebar / header	UC-APP-01	AC-APP-02
REQ-APP-05	FR-APP-06	Application Settings	UC-APP-01	AC-APP-04
REQ-APP-06	FR-APP-07	Application Settings	UC-APP-01	AC-APP-04
REQ-APP-07	FR-APP-08	All "today" calculations	UC-APP-02	AC-APP-05
REQ-APP-08	FR-APP-09	All "today" calculations	UC-APP-02	AC-APP-06
REQ-APP-09	FR-APP-10	Application Settings	UC-APP-01	AC-APP-07
REQ-APP-10	FR-APP-11	Create Task	UC-APP-01	AC-APP-07
REQ-APP-11	FR-APP-12	Application Settings	UC-APP-03	AC-APP-08
REQ-APP-12	FR-APP-13	Notification Center	UC-APP-03	AC-APP-08
REQ-APP-13	FR-APP-14	Application Settings	UC-APP-01	AC-APP-09

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BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-APP-14	FR-APP-15	Application Settings	UC-APP-01	AC-APP-03
REQ-APP-15	FR-APP-16	Application Settings	UC-APP-01	AC-APP-10
REQ-APP-16	FR-APP-17	Application Settings	UC-APP-02	AC-APP-11
REQ-APP-17	FR-APP-18	Sidebar / header, Create Task	UC-APP-01	AC-APP-12
REQ-APP-18	FR-APP-19	Sidebar / header	–	AC-APP-12
REQ-APP-19	FR-APP-20	Application Settings	UC-APP-01	AC-APP-13
REQ-APP-20	FR-APP-21	Application Settings	UC-APP-01	AC-APP-13
REQ-APP-21	FR-APP-22	Application Settings	–	AC-APP-14
REQ-APP-22	FR-APP-23	Application Settings	–	AC-APP-14
BR-APP-01	FR-APP-12	Application Settings	UC-APP-03	AC-APP-08
BR-APP-02	FR-APP-08	All "today" calculations	UC-APP-02	AC-APP-05
BR-APP-03	FR-APP-16	Application Settings	UC-APP-01	AC-APP-10
BR-APP-04	FR-APP-18, FR-APP-23	Application Settings	UC-APP-01	AC-APP-12, AC-APP-14
BR-APP-05	FR-APP-13	Notification Center	UC-APP-03	AC-APP-08
BR-APP-06	FR-APP-14	Application Settings	UC-APP-01	AC-APP-09

Untraced BRD items: **0**.

12. Approvals

Role	Name	Decision	Date
Product owner	Jojo George	Pending	–
Reviewer	Jojo George	Pending	–

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